

How long until every letter returns?

Name: _____

Date: _____

A substitution key uses A, B, C, D, E once each as inputs and once each as outputs. Apply the key to every letter **simultaneously**. Repeat the same key each turn.

Input	A	B	C	D	E
Output	B	C	A	E	D

1. Test this machine.

Start with ABCDE. Record successive messages until the **entire** message first returns. Draw the arrow loops separately.

Turn	Message	Turn	Message
0	ABCDE	4	
1		5	
2		6	
3		7	

The loops, and why their clocks must agree:

2. Make it slower.

Can a different five-letter key take longer to return? A five-letter loop is one candidate. Try several keys; keep a record of the loop sizes.

My best key, its loops, and its return time:

Try it with objects. Explain by pointing, talking, or drawing.

Prove the best is best

3. Replace a huge list by a small one.

Cover the table first. Arrange your five letter cards into loops, working by largest loop: 5, then 4, then 3, then 2, then 1. List how the remaining cards can split without a larger loop. Uncover and compare. A fixed letter is a loop of size 1; rearranging sizes gives no new case.

Loop sizes	First common return time
5	
4 + 1	
3 + 2	
3 + 1 + 1	
2 + 2 + 1	
2 + 1 + 1 + 1	
1 + 1 + 1 + 1 + 1	

4. Explain the classification.

Why does following arrows eventually bring a letter back to **itself**, rather than merge into another loop? Where does using each output once matter? Why does your largest-loop list omit no possible split?

Why all keys are covered, and the largest return time:

A useful language: A loop of size 3 returns on turns 3, 6, 9, The whole key returns at the first turn that is a multiple of *every* loop size. Mathematicians call this the least common multiple.

Try it with objects. Explain by pointing, talking, or drawing.

Does the order of two keys matter?

Key P swaps A and B and leaves C, D, E alone. Key Q swaps B and C and leaves A, D, E alone. “P then Q” means apply P first, then Q to its output.

5. Compare two pipelines.

Start with ABCDE each time. Try P then Q. Reset and try Q then P. Do they agree? One differing letter is enough to disprove agreement.

	After first key	After second key
P then Q		
Q then P		

An arrow picture explaining the difference:

6. Repeat the entire pipeline.

P alone returns in two turns; so does Q. If a turn is now **P then Q**, how many turns until everything returns? Find its loops.

My answer and a loop proof:

Go further: Make two different nonidentity keys whose order *does not* matter. What if they move completely separate sets of letters? Can you undo P then Q using the same two keys?

Try it with objects. Explain by pointing, talking, or drawing.